ORDER-TO-CASH AI DENIAL REDUCTION · 2026

Order-to-Cash

AI Denial Reduction

Everything you need to understand, explain, and confidently talk about the Order-to-Cash AI system — written in plain English with no jargon.

01 What Is This All About?

The big picture — one sentence, one analogy

> *"A lab does blood tests. After the test, they send a bill to the insurance company. Sometimes the insurance company rejects the bill. That rejection is called a denial. This system uses AI to stop denials from happening — before the test is even run."*

■ Think of it like a pizza restaurant

Imagine a restaurant that makes the pizza first, then finds out the customer's credit card is invalid. The pizza is wasted, the oven time is wasted, and the money is gone. Labs have the exact same problem. They run the expensive blood test first, then send the bill to insurance, and only then discover something was wrong with the paperwork from day one. By then it is too late. The money is gone. This system checks the credit card before the pizza goes in the oven.

The gap between when the mistake happens and when it is discovered is weeks. That gap is the entire problem this system solves. The AI finds the mistake at the moment it is born — when fixing it costs nothing — instead of weeks later when the only option is expensive rework or permanent loss.

02 The Core Problem

Why labs lose money — and why it keeps happening

The Problem

A lab order travels through seven stages before payment arrives. Each stage is handled by a completely different team — the doctor's office, the insurance desk, the courier, the lab technician, the billing team. None of these teams can see what the others are doing. When something goes wrong at Stage 1, nobody at Stage 5 knows about it. The problem travels invisibly through all seven stages until the insurance company rejects the claim — and by then the test is done and the money is spent.

The Root Cause

> **Denials are not caused by careless billing.**
>
> They are caused by information arriving too late. The data needed to prevent every denial already exists somewhere in the system — in the doctor's order, in the insurance records, in the claim fields. The problem is that no system today monitors all of it in real time and acts on it before the damage is done.

Three Types of Denial Loss

■ Preventable write-offs

Denials that should never have happened — wrong code, missing permission. Pure waste. The test was valid, the patient was covered, but the paperwork was wrong.

■ Missed deadlines

Every insurance company has a deadline for submitting a claim. Miss it by one day and the money is gone permanently — no appeal allowed, no second chance.

■ Unworked appeals

When a claim is denied, it can often be overturned with a good appeal letter. But writing appeal letters takes time, and there are too many denials for the team to keep up. Money that could be recovered is simply abandoned.

03 The 7 Stages Explained

What happens between a doctor's order and cash in the bank

Think of a lab order like a package being delivered. It goes through seven checkpoints before it arrives as payment. At each checkpoint, something can go wrong — and today, nobody catches it until it is too late.

STAGE 1

Order Entry

What goes wrong today: Doctor sends the lab order. Wrong diagnosis code or missing insurance permission. Nobody checks.

How AI fixes it: Order Agent checks every field instantly — like a spell checker for billing.

CAUGHT EARLY

STAGE 2

Coverage Check

What goes wrong today: Lab checks if insurance covers this test. Insurance permission expires quietly. Nobody watches.

How AI fixes it: Coverage Agent monitors every permission around the clock. Auto-renews before expiry.

AUTH SECURED

STAGE 3

Specimen Transport

What goes wrong today: Blood sample travels to the lab. Gets too warm during delivery. Becomes unusable — test unrunnable.

How AI fixes it: Logistics Agent reads IoT sensor data from courier systems in real time. Flags temperature breaches while re-collection is still possible. See Section 05 for full detail.

SAMPLE SAFE

STAGE 4

Lab Processing

What goes wrong today: The test runs. Takes longer than expected. Insurance billing deadline silently passes.

How AI fixes it: Lab Agent watches every deadline. Pushes the claim to the front of the billing queue the moment results are ready.

FILED ON TIME

STAGE 5

Claim Submission

What goes wrong today: Lab sends the bill. Wrong billing code or routing error. High chance of rejection.

How AI fixes it: Billing Agent scores rejection risk for every claim using a trained ML model. Automatically fixes it before it goes out. See Section 06 for detail on the model.

CLEAN CLAIM

STAGE 6

Insurance Decision

What goes wrong today: Insurance approves or rejects. Rejected claims could be appealed but team is too busy.

How AI fixes it: Appeals Agent reads the insurer's own policy and writes a complete appeal letter in seconds.

APPEAL READY

STAGE 7

Payment

What goes wrong today: The lab has done the test and sent the bill — but the finance team only finds out about denied claims, expired authorizations, and missed deadlines from a weekly report that is already days old. By then the window to fix most problems has already closed.

How AI fixes it: The live AI dashboard replaces the stale weekly report with a real-time view updated every 60 seconds — showing which claims are at risk right now, which authorizations are about to expire, and which payers are behaving differently. The CFO acts before the money is lost, not after.

LIVE VIEW

04 How AI Fixes Each Stage

Three agents cover the full denial lifecycle — prevent, catch, recover

The full system has seven agents — one for each stage. For the demo, three agents are built because they cover the three most important moments: stopping the denial before it is born, catching it before the insurance company sees it, and recovering it after the insurance company has already rejected it.

PREVENT

Order Agent — Stage 1

Stops the denial at its origin

What it checks

Every field in every lab order — the diagnosis code, the billing code, the insurance permission requirement, the doctor's ID format. This runs as pure rules — no AI guessing involved. It is fast, reliable, and catches errors that humans miss because there are too many orders to check manually.

When AI gets involved

If the doctor wrote a clinical note — like 'patient showing signs of hereditary cancer risk' — the AI reads that note and suggests the correct diagnosis code, citing the exact medical coding guideline that supports it. A human rule then checks whether that suggestion is valid before anything happens.

Why this stage matters most

A mistake caught here costs zero. No test has been run. No sample has been collected. The same mistake caught at Stage 5 means a denial, rework, and delayed payment. The same mistake missed entirely becomes a permanent write-off.

CATCH

Billing Agent — Stage 5

Stops the denial before the insurance company sees it

What it checks

Every claim gets a denial risk score before it leaves the system. The score is calculated by a machine learning model trained on your own historical claim outcomes — it knows which billing codes, which insurance companies, and which combinations have historically caused rejections. Full detail in Section 06.

What happens when risk is high

The system automatically corrects the claim — fixes the billing modifier, adjusts the routing, applies the bundling rules — and then rescores it. Both scores are shown: what it was before the fix and what it became after. The claim goes out clean. The insurance company never sees the flawed version.

Why this is safe

No AI is making the billing decision. A statistical model identifies the risk. A rules engine applies the correction. The same inputs always produce the same output — fully auditable, fully explainable.

RECOVER

Appeals Agent — Stage 6

Recovers money from denials that already happened

What happens when a claim is denied

The denial reason code — a standard insurance rejection code — is mapped to an appeal strategy using a lookup table. This part is pure rules, no AI. Then the AI reads the insurance company's own coverage policy document and drafts a complete appeal letter citing the exact section that supports payment.

What the human auditor does

The appeal letter appears in the HITL dashboard. They see the policy citation, the AI confidence score, and the complete decision trail. They approve it in one click. The entire process takes three minutes instead of twenty-five.

Why recoverable denials are currently abandoned

Today, writing one appeal letter takes a billing specialist significant time. With hundreds of denials per month, there is not enough time to appeal all of them. The Appeals Agent removes that constraint — every recoverable denial gets a properly documented appeal, every time.

05 How 3 Agents Cover All 7 Stages

Why the POC is scoped to 3 agents — and how they address the full cycle

A common question: the system has 7 stages but only 3 agents in the demo. Does that mean 4 stages are unaddressed? No — here is exactly how the 3 agents map across all 7 stages.

Stage 1 — Order Entry

■ Fully covered — Agent 1 primary job

- The Order Agent is built specifically for this stage.
- Every order is validated before the specimen is collected — wrong diagnosis codes, missing prior auth, invalid billing pairs.
- Fixing a problem here costs zero. No test has been run. No money has been spent.

Stage 2 — Coverage Check

■ Partially covered — Order Agent side effect

- When the Order Agent checks prior auth requirements against the payer rules table, it is performing a coverage check at the same time.
- It catches the most common coverage failure — missing prior authorization — before collection begins.
- A full real-time payer eligibility API check is included in the coverage verification step.
- For the demo this is sufficient because the demo trigger for Stage 2 is a missing prior auth — which the Order Agent catches.

Stage 3 — Specimen Transport

■■ Out of POC scope — requires IoT courier integration

- Temperature tracking requires IoT sensor integration with courier systems.
- This is a real hardware-plus-software integration that cannot be built in two days.
- Specimen tracking requires IoT sensor integration with courier systems — out of scope for a two-day hackathon build.
- This does not weaken the demo — the three chosen demo triggers have higher financial impact than a temperature breach.

Stage 4 — Lab Processing

■■ Out of POC scope — requires Lab Information System integration

- Filing deadline monitoring requires live integration with the Lab Information System.
- That integration is out of scope for a hackathon POC.
- Indirect coverage only: the Billing Agent in Stage 5 checks the timely filing window as part of its pre-submission scrub.

Stage 5 **Claim Submission** ■ **Fully covered — Agent 2 primary job**

- The Billing/Denial Agent is built specifically for this stage.
- Every claim is scored for denial risk before it leaves the system.
- High-risk claims are automatically corrected and rescored — both scores visible on screen simultaneously.
- This is the most visually demonstrable agent in the entire demo.

Stage 6 **Insurance Decision** ■ **Fully covered — Agent 3 primary job**

- The Appeals Agent is built specifically for this stage.
- Denied claims are caught, the payer policy is retrieved via RAG, and a complete appeal letter is drafted with citation in seconds.
- The human auditor approves in one click via the HITL dashboard.
- This closes the full denial lifecycle — prevent at Stage 1, catch at Stage 5, recover at Stage 6.

Stage 7 **Payment Dashboard** ■ **Indirectly covered — all 3 agents feed the dashboard**

- Every action the three agents take writes to the Governance Sink and updates the Gold layer in BigQuery.
- The live dashboard reads from that Gold layer in real time — no additional agent needed for the demo.
- When the Order Agent catches a prior auth issue, the dashboard shows it as a prevented denial immediately.
- When the Billing Agent rescores a claim, the revenue risk score updates on the dashboard in seconds.
- Stage 7 is fully demonstrable in this POC through the data the three agents already generate.

> *"Three agents that work perfectly cover the three highest-impact stages and demonstrate every architectural principle — prevent, catch, recover, audit. Stages 3 and 4 require physical hardware integrations out of scope for this build. A focused demo that works beats an ambitious demo that breaks."*

06 The Denial Prediction Model

How it is trained on your own historical claims — and why that matters

This section explains exactly what the denial prediction model is, what data it needs to be trained, and why that data already exists in your billing system today. No separate ML platform is required — the entire training process runs inside BigQuery.

Why the Model Must Be Trained on Your Own Data

> **Denial patterns are specific to you — not universal.**
>
> A model trained on generic healthcare billing data would know that 'claims sometimes get denied.' That is not useful. What you need to know is: which specific billing codes, submitted to which specific insurance companies, under which specific diagnosis combinations, get denied — for your lab, in your market, with your payer contracts. This is specific to your organisation. Your historical claims are the only source of truth for it.

What Makes Denial Prediction Specific to Each Lab

■ Each payer behaves differently

Payer A may deny molecular genetic tests at a high rate but approve chemistry panels freely. Payer B may be the opposite. A pre-trained model trained on a national dataset would average these out and give you misleading predictions for both. Your model needs to learn the specific behaviour of your specific payer mix.

■ Each test menu is different

A lab that specialises in oncology testing has a completely different denial profile from a lab that primarily does routine blood panels. The patterns that predict denial for a BRCA genetic panel are nothing like the patterns for a basic metabolic panel. The model must be trained on the test types your lab actually performs.

■ Billing rules change over time

Insurance companies update their coverage policies, prior auth requirements, and modifier rules regularly. A model trained two years ago on general data will not reflect the current rules for your current payer contracts. Your model, trained on your recent claims, reflects today's reality.

■ Geography matters

State Medicaid programs have different rules than federal Medicare. Regional commercial payers have local policies. A model trained on national data blends all of this together and loses the specific patterns that predict denials in your region.

How the Model Is Actually Trained

The good news: this is not complicated machine learning. It uses a logistic regression model — one of the simplest, most reliable, and most interpretable statistical techniques available. Here is what it requires:

Input needed	What it is	Do you already have it?
Historical claims	Claims submitted to payers over the past one to two years, with their outcomes — paid, denied, or adjusted	**Yes — this lives in your billing system today. It is the data you already generate every day.**
Denial reason codes	The reason the payer gave for each denial — these are standard industry codes like CO-4, CO-16, CO-96	**Yes — these are recorded with every denial in your billing system.**
Claim features	The fields on each claim: CPT code, ICD-10 code, payer ID, modifier, NPI, place of service, date of service	**Yes — these are on every claim you have ever submitted.**
BigQuery ML	The Google Cloud tool that trains the model directly on the data in BigQuery — no separate ML platform needed	**Already in the stack — no additional tool required.**

What the Training Process Looks Like

Training the denial prediction model is not a months-long AI project. It is a well-understood data science task that follows a clear process:

Step 1 — Export historical claims from your billing system: Pull the last 12 to 24 months of submitted claims with their outcomes. This data already exists — you are just moving it into BigQuery.

Step 2 — Label each claim: Each claim is already labelled by outcome: paid, denied, adjusted. The model learns what a denied claim looks like compared to a paid claim.

Step 3 — Train the model in BigQuery ML: A single SQL command in BigQuery ML trains a logistic regression model on your historical data. This takes minutes, not days. No Python, no notebooks, no separate ML environment required.

Step 4 — Validate on holdout data: A portion of the historical claims is held back during training to test accuracy. The model's performance is measured on claims it has never seen — this is how you know it will work on new claims too.

Step 5 — Deploy and monitor: The trained model scores every new claim before submission. As new claims are processed and outcomes arrive, the model can be retrained periodically — monthly or quarterly — to stay current with payer behaviour changes.

> *"The data you need to train this model already exists in your billing system today. You do not need to buy anything new or build a separate ML platform. BigQuery ML trains the model directly on your historical claims with a single command."*

07 The Two Golden Rules of the AI

How every decision is made safely in a regulated environment

■■ Rule 1 — The AI reads. The rules decide.

The AI reads messy human documents — doctor's notes written in shorthand, insurance policy PDFs full of legal language, denial letters with technical codes. It extracts clean, structured information from them. Then a separate rules engine takes that clean information and makes the billing decision. The AI never makes the final call. The AI is the translator, the rules engine is the judge. The translator tells the judge what the documents say. The judge decides what to do.

■■ Rule 2 — When the AI is not confident, a human steps in.

Every AI response gets a confidence score — a number that shows how sure the AI is. If the score falls below the safety threshold, the system automatically routes the case to a human auditor for review before any action is taken. This is called HITL — Human in the Loop. The AI never acts alone on something it is unsure about. There is no path for a low-confidence, unreviewed AI output to reach a claim.

Why These Two Rules Matter for Healthcare

In most industries, an AI making an occasional mistake is acceptable. In healthcare billing, a wrong AI decision can trigger a compliance investigation, a HIPAA violation, or a regulatory audit. These two rules mean the system is safe to use in a regulated environment because the AI never has unchecked authority over a billing decision.

> *"The AI is the translator, the rules engine is the judge, and the human auditor is the final check. All three work together. None of them works alone."*

08 The Two Foundation Layers

The Governance Sink and HITL — what makes the system trustworthy

These two layers run underneath all seven stages. They are not part of any specific stage — they support every stage. They are what make the system trustworthy enough to use in a regulated healthcare environment.

The Governance Sink

> **✈■ Think of it as the black box recorder on an aeroplane**
>
> Every commercial flight carries a black box that records everything — every instrument reading, every conversation in the cockpit, every decision made. If something goes wrong, investigators can replay every second of the flight. The Governance Sink is the black box for every billing decision. Every AI action, every rule applied, every document read, every human approval — all of it is logged permanently and can never be deleted or changed.

What it records:

- Which AI agent took the action
- Which rule was applied and which version of that rule
- Which insurance policy document was read and which section
- The AI confidence score at the time of the decision
- Whether a human auditor reviewed and approved the output
- The exact timestamp of every step

> **Why this matters for audits**
>
> Today, if a regulatory auditor asks why a claim was billed a certain way, it takes weeks of document searching and staff time to reconstruct the answer. With the Governance Sink, the answer is a single database query that returns in seconds. This turns a compliance liability into a competitive advantage.

The HITL Dashboard — Human in the Loop

HITL stands for Human in the Loop. It is the screen the clinical auditor uses to review cases where the AI was not confident enough to act alone.

What the auditor sees today	What the auditor sees with HITL

A claim number and a rejection code. Nothing else. They spend significant time searching for the payer policy, understanding why it was denied, and writing a justification from scratch.	The payer policy section already retrieved with the relevant passage highlighted. The AI recommendation. The confidence score. The complete decision trail. Everything in one screen. Decision made in minutes.

Every correction an auditor makes is automatically logged as a training example — so the AI learns from every human override and gets more accurate over time for that specific lab's payer mix.

09 The Tools — Plain English

What each tool actually does — no technical background needed

Every tool has one specific job in the system. Here is what each one does in plain language.

Tool	Plain English Name	What It Actually Does
■ Gemini 2.5	**Google's AI brain**	Reads doctor's notes and insurance policy documents. Extracts the important information. Writes appeal letters. It never makes the final billing decision — it just reads and translates.
■ Vertex AI Search	**The policy library with a confidence score**	Finds the exact section of an insurance policy document that applies to a claim. Also gives a confidence score — a rating of how sure it is about what it found. If the score is too low, a human steps in.
■■ BigQuery	**The memory that never forgets**	The database that stores everything — every order, every claim, every AI decision, every audit log. Nothing can be deleted or changed. Every fact can be found in seconds.
■ BigQuery ML	**The denial predictor**	A statistical model trained on your own historical claims. Scores the risk of a claim being rejected before it is sent. Every prediction is explainable — you can always see why a claim was flagged. See Section 06 for full detail on how this is trained.
■ LangGraph	**The coordinator**	Tells each AI agent when to act, what to pass to the next agent, and when to escalate to a human. Think of it as the air traffic controller for the whole system.
■■ FastAPI Gateway	**The security checkpoint**	Every AI request must pass through this door. Patient data is removed before anything reaches the AI. Every response is checked. Every interaction is logged. Nothing bypasses it.
✂■ Cloud DLP	**The patient data eraser**	Automatically finds and removes patient names, dates of birth, and member IDs from every request before it reaches the AI. Patients are never identified inside the AI system.
■■ Cloud Run	**The engine room**	Where the system lives and runs. Automatically handles more load when it gets busy and scales back when it gets quiet. No infrastructure team needed during a demo.

Tool	Plain English Name	What It Actually Does
■ Looker or Grafana	**The dashboard**	The screen the CFO and auditor use. Shows live revenue risk, denial trends, and the HITL claim queue. Looker is the natural choice if your organisation is already on GCP — native BigQuery integration, no ETL, self-serve for business users without SQL. Grafana is an equally strong alternative if your organisation already uses it for monitoring — it handles live time-series data excellently, connects to BigQuery via plugin, and is open-source with no additional licensing cost. Either works architecturally. Choose the one your team already knows.
■ FHIR R4	**The universal language**	The standard format that hospitals, clinics, and labs all use to share patient orders. Using it means this system works with any hospital software — Epic, Cerner, Oracle — without custom integration.

10 The Demo Story

Five steps — five minutes — one complete story

When showing this system to anyone — a manager, an executive, a judge — the demo tells one story in five beats. No slides during the demo itself. Just the system running live.

PREVENT

What happens: An order arrives with a missing prior authorization for an expensive test.

What the system does: The Order Agent flags it instantly. The system automatically generates the authorization request using the insurer's own coverage policy — citing the exact policy section. The denial is stopped before the needle touches the patient.

CATCH

What happens: A claim is built for a test with a high chance of being rejected.

What the system does: The Billing Agent scores the denial risk as high. It applies the correct billing modifier. It rescores the claim to low risk. Both scores are visible on screen at the same time. The claim goes out clean. The insurer never sees the flawed version.

RECOVER

What happens: An intentional denied claim enters the Appeals Agent.

What the system does: The system retrieves the insurer's coverage policy via AI search. Gemini drafts a complete appeal letter with the specific policy section cited. The confidence score is displayed on screen. The letter appears in the HITL queue.

GOVERN

What happens: The clinical auditor opens the HITL dashboard.

What the system does: They see the policy citation, the confidence score, and the full decision trail in one view. They approve the appeal letter in one click. The approval is logged to the Governance Sink.

CLOSE

What happens: A live database query runs on screen.

What the system does: The query returns the complete history of every AI action for that claim — every rule applied, every document read, every decision made — in seconds. Complete. Immutable. Instantly retrievable. This is the moment that wins the room.

> *"Every AI decision in this system is reproducible, citable, and auditable to the specific payer policy that grounded it — because in healthcare, the AI decided is not a defensible answer. Ours always is."*

11 Stage 7 Deep Dive — Payment

The most misunderstood stage — explained completely

Most people look at Stage 7 and think — it is just a dashboard, nothing exciting. This section explains why that thinking is wrong, and why Stage 7 is actually what makes every other stage meaningful.

First — What Does 'Revenue at Risk' Mean?

> **A simple definition**
>
> When a lab performs a test, they expect to get paid. But between performing the test and receiving the money, many things can go wrong — a prior auth is expiring, a claim has a high chance of being rejected, an appeal deadline is approaching. All of that money — earned but not yet safely collected — is called revenue at risk.

The Problem With Today's Reporting

Today the CFO opens a revenue report every Monday morning. That report tells them what happened last week. Three things are wrong with this:

Problem 1 — The report is already old

It shows what happened last week. Not what is going wrong right now. By the time you read about a problem, it has been a problem for days.

Problem 2 — You only see what already went wrong

The report shows denials that already happened. It does not show which claims are about to be denied, which authorizations are about to expire, or which deadlines are three days away. You are always looking backwards.

Problem 3 — You cannot act in time

By the time the problem appears in the report, the window to fix it is often closed. The filing deadline passed. The authorization expired. The appeal window closed. The money is gone.

> **■ Bank statement in the mail vs. banking app on your phone**
>
> Getting a weekly revenue report is like receiving your bank statement in the mail once a week. You open it on Monday and find out your account went negative last Tuesday. Too late. The AI dashboard is like having a banking app. You get a notification the moment a problem emerges — while you still have time to act and fix it. Same information. Completely different timing. Timing is everything.

What the Live Dashboard Shows

■ **Revenue at risk right now**

Every claim in progress, scored by how likely it is to result in payment or loss. A high-value test with an expiring authorization shows up as high risk — right now, not next week.

■ **Which insurers are causing the most risk**

If a specific insurer suddenly starts rejecting more claims than usual, the pattern shows up in real time. The CFO can act today instead of discovering the trend next month.

■ **Which test types are most at risk**

If genetic tests are approaching filing deadlines because lab processing is running long, that is visible immediately — not after the deadlines have passed.

■ **Where the other agents are intervening**

Every time the Order Agent catches an error, every time the Billing Agent fixes a claim, every time the Appeals Agent drafts a letter — all of it feeds into the dashboard as a live record.

Why Stage 7 Makes Every Other Stage Meaningful

> **The executive dashboard is the proof layer**
>
> Without Stage 7, the CFO has no way of knowing whether the system is working. They cannot see how many denials are being prevented, which insurers are improving, or whether the AI is actually saving money. With Stage 7, the CFO can point to the dashboard and say: these are the claims that would have been denied, these are the authorizations that would have expired, this is the revenue that was protected. That visibility is what turns an AI project into a business decision that gets funded.

> *"Stage 7 changes the CFO from someone who finds out about financial problems after they happen, into someone who sees financial risk before it becomes financial loss — and can act on it while there is still time."*

The One Sentence to Remember

> *"Every denial has a root cause. This is the only system that finds it before the money is spent."*